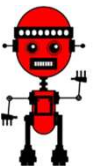


CONTENT

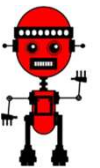
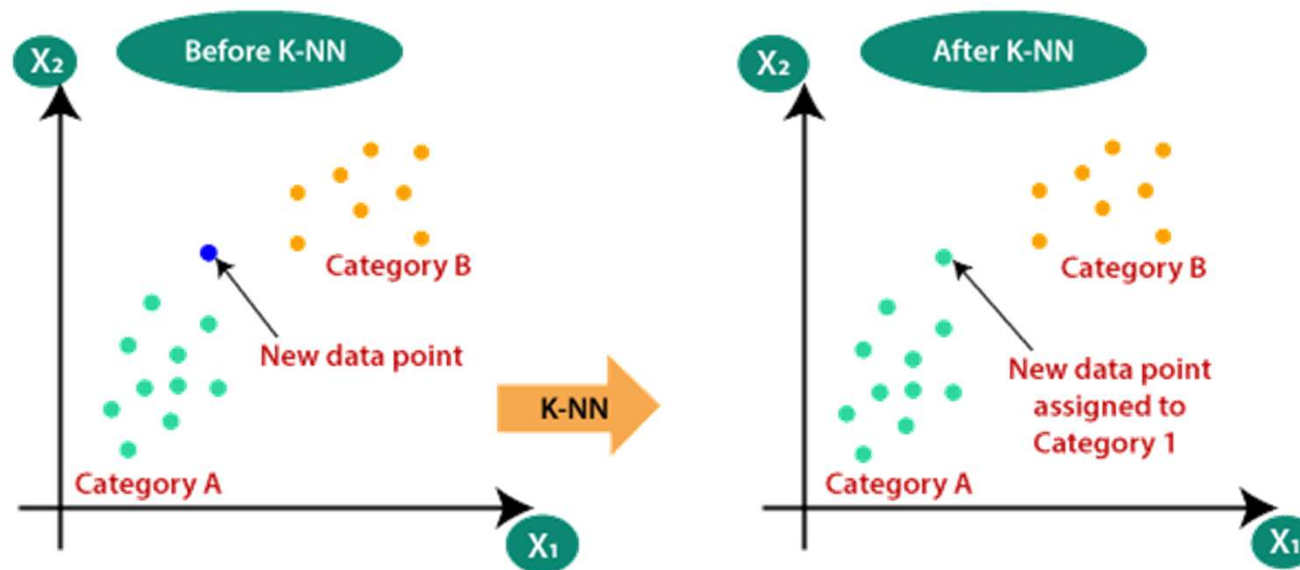


- Concept of KNN
- How does it work?
- Selection of K parameter
- Use Cases
- Merits and Demerits



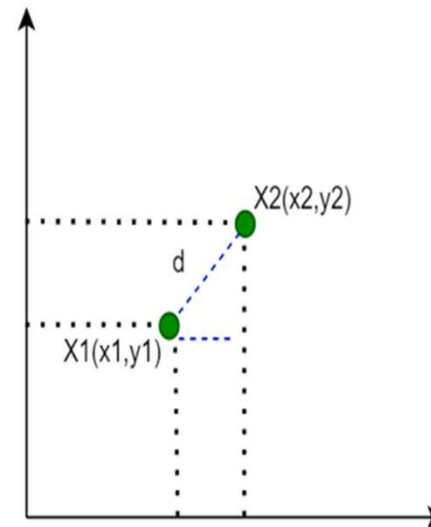
Goal of KNN

Find nearest neighbors of a given query data point



Concept of KNN

- It is Supervised machine learning algorithm
- K-Nearest Neighbour
- K defines number of considered nearest neighbour
- Distance based machine learning algorithm
- Non-parametric algorithm
- lazy learner algorithm



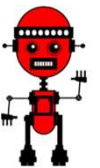
$$d = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

Distance Metrics

Required for formation of decision boundary

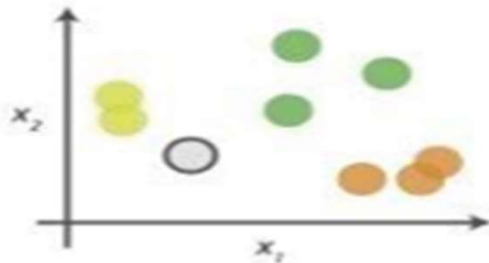
Few preferred distance measures are :

- Euclidean distance
- Manhattan distance
- Minkowski distance
- Hamming distance



How does KNN work?

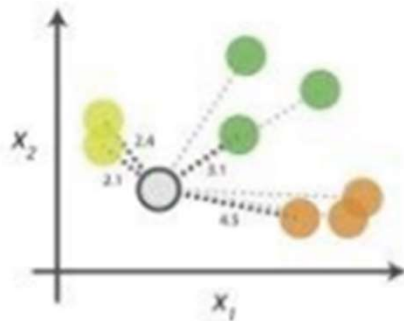
0. Look at the data



2. Find neighbours

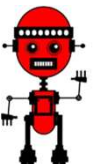
Point Distance			
		2.1	→ 1st NN
		2.4	→ 2nd NN
		3.1	→ 3rd NN
		4.5	→ 4th NN

1. Calculate distances



3. Vote on labels

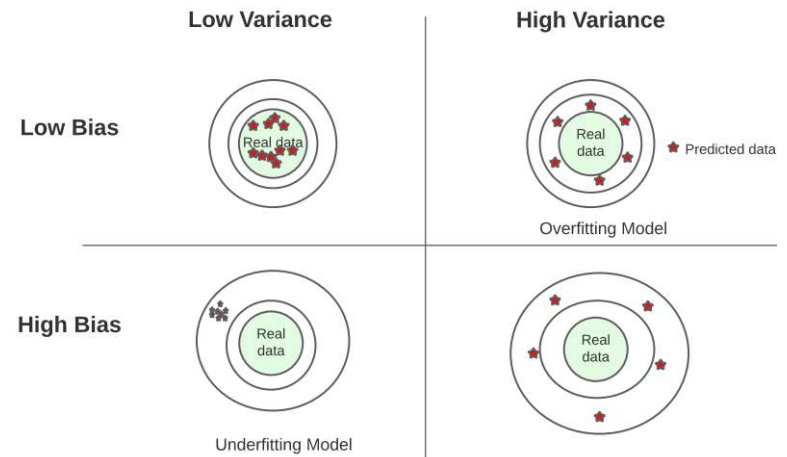
Class	# of votes	
	2	→ Class  wins the vote! Point  is therefore predicted to be of class  .
	1	
	1	



Selection of K - Value

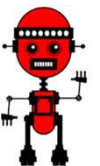
- K - value defines, how many neighbours will be checked to determine the classification of a specific query point.
- Lower values of k can have high variance, but low bias, and larger values of k may lead to high bias and lower variance
- Select odd number for k to avoid ties in classification
- Selection of optimal value of K is required otherwise it can lead to overfitting and underfitting case.

Bias and Variance are prediction error.



Bias – Difference in between predicted value and actual value.

Variance – Amount of variation in the prediction when model is trained with different training set.



Use cases of KNN in real life

- Recommendation Engines
- Finance
- Healthcare
- Pattern Recognition

